

06 DSS Design and User Guide

Disclosure: all results use the synthetic `fra_synthetic_claims` dataset. Scores may reflect generator rules. Real-world performance is unknown. This demo must not drive real claim decisions.

1. Architecture

`fra_dss.dss.engine.recommend(claim)` combines: calibrated-style probabilities, SHAP sentences, optional rejection reasons, what-if on actionable fields only, ETA on resolved-trained regressor, `scheme_rules` from the Flask backend, and audit flags.

Triage score = $0.45 \cdot P(\text{reject}) + 0.35 \cdot \text{delay_norm} + 0.20 \cdot \text{age_norm}$ (`configs/dss_weights.yaml`).

2. How to run

`streamlit run ml/app/streamlit_app.py` inside `ml/.venv`. Flask additive routes: `/api/dss/benchmark`, `/reasons`, `/what-if`, `/eta`, `/triage`.

ETA champion: `gbr`. Rejection-reason champion: `random_forest`.